Subject: **Recycling of Materials (OE – 5)**								
Program: **B. Tech in Metallurgical Engineering**				Subject Code: **MME0406**			Semester: I**V**	
Teaching Scheme (Hours per week)				**Examination Evaluation Scheme (Marks)**				
Lecture	**Tutorial**	**Practical**	**Credits**	**University Theory Examination**	**University Practical Examination**	**Continuous Internal Evaluation (CIE)-Theory**	**Continuous Internal Evaluation (CIE)-Practical**	**Total**
3	0	0	3	16/40	0	24/60	0	100

Course Objectives

1. To make the students aware about the serious issue and how to combat the problem.
2. Classifications of the waste materials, recycling methods, recycling symbol and the technology of recycling within the economic barriers will be dealt with in this course.
3. Students will learn how to make public conscious of recycling of wastes and economic consideration.

CONTENT

UNIT-I

[10 hours]

Definition of terms, recycling and recovery.

Importance, purpose, and contribution of economy and human and environmental health of recycling. Recycling symbols, Recyclable wastes. Recycling statistics in India and World.

UNIT-II

[10 hours]

Defining the stages of recycling processes.

Collection, Separation and recycling stages of plastic wastes, steel/stainless steel, aluminum, and copper wastes, and recycling symbols.

UNIT-III

[10 hours]

Collection, Separation and recycling stages of composite wastes, glass wastes, building wastes, and recycling symbols.

UNIT-IV

[10 hours]

Collection, Separation and recycling stages of accumulator and battery wastes, electronic wastes, and recycling symbols.

Collection, transportation and disposal of harmful wastes.

Course Outcomes

1. The students will be made familiar with different types of waste materials and their disposal problems. Thus, necessitating the knowledge on recycling of the waste materials.
2. Distinguish the recycling, recovery, reuse, reduce and disposal of wastes.
3. Comprehend the recycling symbols.
4. The students will be able to know about the collection and separation systems of wastes.
5. The students will be able to analyses awareness studies to public.

Reference Books

1. Material Recycling – Trends and perspectives, Edited by Dimitris S. Achilias, InTech, 2012.